



AMITY UNIVERSITY

**AMITY UNIVERSITY ONLINE, NOIDA,
UTTAR PRADESH**

In partial fulfilment of the requirement for the award
of degree of **Bachelor of Computer Application**
(Data Analysis)

(Discipline - IT/EVS/PPG/etc.)

**TITLE: FLIGHT PRICE PREDICTION USING LINEAR
REGRESSION**

Guide Details:

Name: Shitiz Goel

Designation: Associate Data Scientist

Submitted By:

Name of the Student - Subash Dey

Enrolment. No: A9922523002580

ABSTRACT

The aviation industry has experienced rapid growth over the past few years, leading to an increase in the number of passengers traveling by air. Along with this growth, flight ticket prices have become highly dynamic and change frequently based on several factors such as airline company, source and destination cities, duration of the journey, number of stops, departure and arrival timings, class type, and the number of days between booking date and journey date¹. Because of the constant price fluctuations, passengers often face difficulty in deciding the best time to book tickets at affordable prices. Predicting flight ticket prices has therefore become an important area of research in the fields of Data Science and Machine Learning.

The main objective of this project is to develop a Flight Price Prediction model using Machine Learning techniques. The project aims to analyze historical flight data and predict ticket prices based on various influencing factors. The dataset used for this project was collected from Kaggle and contains important flight-related attributes such as airline name, source city, destination city, departure time, arrival time, duration, class type, number of stops, days left before journey, and ticket price. These features play a significant role in determining the overall price of a flight ticket.

The project begins with data collection and pre-processing. During the pre-processing stage, unnecessary columns are removed, missing values are treated, duplicate records are identified, and data types are corrected wherever necessary. Data pre-processing is an important step because clean and properly formatted data helps improve the accuracy and efficiency of prediction models.

After pre-processing, Exploratory Data Analysis (EDA) is performed to better understand the dataset and identify patterns, trends, and relationships among variables. Different visualization techniques such as histograms, boxplots, scatter plots, and heatmaps are used to analyse the distribution of features and detect outliers. The analysis shows that ticket prices are influenced by several important factors. For example, Business class tickets are significantly more expensive than Economy class tickets, flights with longer duration tend to have higher prices, and ticket prices usually increase as the departure date approaches. The analysis also reveals that airlines, number of stops, travel class and departure timings have a strong impact on pricing patterns.

Once the dataset is analysed, machine learning algorithms are applied to build prediction models. Regression algorithms are commonly used in this type of project because the target variable, ticket price, is continuous in nature. Different machine learning models can be used, such as Linear Regression, Decision Tree Regression, Random Forest Regression, and other advanced algorithms. These models learn patterns from historical flight data and attempt to predict future ticket prices with high accuracy. In this project, we used the Linear Regression Model because it provides better explainability over other machine learning models.

The dataset is divided into training data and testing data before model training. The training data is used to teach the machine learning model how different features affect ticket prices, while the testing data is used to evaluate the model's performance on unseen data. Evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score are used to measure the prediction accuracy and overall performance of the models.

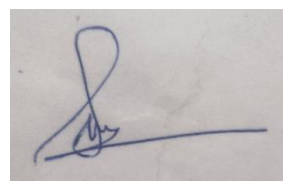
The Flight Price Prediction project demonstrates the practical application of Data Analytics and Machine Learning in solving real-world problems. The project not only helps passengers estimate ticket prices but also helps airline companies and travel agencies understand customer booking behavior and pricing trends. Accurate prediction systems can help users make smarter booking decisions and save money by identifying the best time to purchase tickets.

This project also provides valuable learning experience in different stages of the Data Science workflow, including data pre-processing, exploratory data analysis, feature engineering, model training, testing, and evaluation. It improves understanding of how machine learning algorithms work on real-world datasets and how data-driven approaches can be used to make intelligent predictions.

In conclusion, the Flight Price Prediction project successfully highlights the importance of Machine Learning in the aviation industry. By analyzing historical flight data and identifying important pricing factors, the system can predict flight ticket prices with reasonable accuracy. The project demonstrates how technology and data analysis can help solve practical business problems and improve decision-making processes for both customers and airline service providers.

DECLARATION

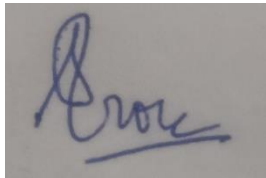
I, **Subash Dey**, a student pursuing **BCA, 6th Semester** at Amity University Online, hereby declare that the project work entitled “**Flight Price Prediction using Linear Regression**” has been prepared by me during the academic year 2026 under the guidance of Shitiz Goel, Associate Data Scientist, Indian Institute of Technology, Roorkee. I assert that this project is a piece of original bona-fide work done by me. It is the outcome of my own effort and that it has not been submitted to any other university for the award of any degree.

A handwritten signature in blue ink, consisting of a stylized 'S' followed by a horizontal line and a small flourish.

Signature of Student

CERTIFICATE

This is to certify that **Subash Dey** of Amity University Online has carried out the project work presented in this project report entitled “**Flight Price Prediction Using Linear Regression**” for the award of **Bachelor of Computer Application** in **Data Analysis** under my guidance. The project report embodies results of original work, and studies are carried out by the student himself. Certified further, that to the best of my knowledge the work reported herein does not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.



Signature

Shitiz Goel

Associate Data Scientist

TABLE OF CONTENTS

<u>TABLE OF CONTENTS</u>		
ABSTRACT		2-4
CANDIDATE’S DECLARATION		5
CERTIFICATE		6
Chapter 1	INTRODUCTION OF THE TOPIC	11-14
Chapter 2	REVIEW OF LITERATURE	15-19
Chapter 3	RESEARCH OBJECTIVES AND METHODOLOGY	20-22
Chapter 4	DATA ANALYSIS AND RESULTS	23-70
Chapter 5	FINDINGS AND CONCLUSION	71-73
Chapter 6	RECOMMENDATIONS AND LIMITATIONS	74
REFERENCE		75

LIST OF TABLES

Sl. No:	Table and Name	Page No.
1	Table 4.1: One-Hot Encoding of Airline Feature	54
2	Table 4.2: Statistical Summary of OLS Regression Model	62
3	Table 4.3: Airline-wise Effect on Flight Prices	65
4	Table 4.4: Impact of Departure Time on Ticket Prices	65

LIST OF FIGURES

Sl. No:	Figure and Name	Page No.
1	Figure 4.1: Distribution of airline	23
2	Figure 4.2: Distribution of Source City	24
3	Figure 4.3: Distribution of Destination City	25
4	Figure 4.4: Distribution of Stops	26
5	Figure 4.5: Distribution of Arrival Time	27
6	Figure 4.6: Distribution of Departure Time	28
7	Figure 4.7: Distribution of Class	29
8	Figure 4.8: Distribution of Duration	31
9	Figure 4.9: Distribution of Days Left	32
10	Figure 4.10: Distribution of Price	33
11	Figure 4.11: Boxplot of Duration	35
12	Figure 4.12: Boxplot of Days Left	36

13	Figure 4.13: Boxplot of Price	37
14	Figure 4.14: Outlier Detection and Removal using IQR Method	38
15	Figure 4.15: Price Distribution by Airline	40
16	Figure 4.16: Price Distribution by Source City	41
17	Figure 4.17: Price Distribution by Destination City	42
18	Figure 4.18: Price Distribution by Departure Time	43
19	Figure 4.19: Price Distribution by Arrival Time	44
20	Figure 4.20: Price Distribution by Stops	45
21	Figure 4.21: Price Distribution by Class	46
22	Figure 4.22: Duration vs Price	47
23	Figure 4.23: Days Left vs Price	48
24	Figure 4.24: Correlation Heatmap Analysis	49

CHAPTER 1: INTRODUCTION TO THE TOPIC

In today's world, air travel has become one of the most common and fastest ways of transportation. People travel by flights for business, education, tourism, family visits, and many other reasons. However, the price of flight tickets changes frequently based on different factors such as airline company, travel date, departure time, destination, number of stops, and booking time. Sometimes ticket prices increase suddenly, making it difficult for passengers to plan their travel within budget. Because of these changing prices, predicting flight ticket prices has become an important problem in the travel industry. Flight price prediction helps customers estimate future ticket prices and make better booking decisions. It can also help travel companies and airlines understand pricing patterns and improve their services.

Traditional and Modern Methods for Flight Price Prediction

Traditional Methods for Flight Price Prediction

Before the use of Machine Learning and Artificial Intelligence, flight ticket prices were mainly decided using traditional methods. These methods were simple and mostly based on fixed rules and manual analysis.

1. Fixed Pricing Method

In this method, airlines set fixed prices for tickets based on routes, distance, and class type. Prices did not change frequently.

Example:

A flight from Delhi to Mumbai may always cost around ₹5000 in economy class.

2. Seasonal Pricing

Airlines changed ticket prices depending on travel seasons, holidays, or festivals.

Example:

Flight prices increase during Diwali, New Year, or summer vacations because more people travel during these times.

3. Demand-Based Pricing

Prices were adjusted based on passenger demand. If more people booked tickets, prices increased.

Example:

If many passengers book tickets for a weekend flight, the ticket price becomes higher.

4. Manual Analysis

Travel agents and airline companies manually analyzed previous booking data and market trends to estimate prices.

Limitations of Traditional Methods

- Less accurate
- Time-consuming
- Difficult to handle large amounts of data
- Cannot predict sudden market changes properly
- Mostly dependent on human decisions

Modern Methods for Flight Price Prediction

With the growth of technology, airlines and travel companies now use modern techniques such as Data Science, Machine Learning, and Artificial Intelligence to predict flight prices more accurately.

1. Machine Learning Models

Machine Learning algorithms learn patterns from historical flight data and predict future prices automatically.

Common algorithms used:

- Linear Regression
- Decision Tree Regression

- Random Forest Regression
- XGBoost
- Support Vector Machine (SVM)

Example:

The model predicts ticket price based on airline, source city, destination city, travel date, duration, and number of stops.

2. Data Analysis and Visualization

Modern systems use data analysis techniques to understand trends and patterns in flight prices.

Techniques include:

- Exploratory Data Analysis (EDA)
- Univariate Analysis
- Bivariate Analysis
- Correlation Analysis

These methods help identify which factors affect ticket prices the most.

3. Real-Time Dynamic Pricing

Airlines now use real-time systems that continuously update prices based on:

- Demand
- Seat availability
- Weather conditions
- Fuel prices
- Competitor pricing

Example:

If only a few seats are left on a flight, the ticket price automatically increases.

4. Artificial Intelligence (AI)

AI-based systems can analyze huge amounts of travel data and make smarter predictions.

Benefits:

- Better accuracy
- Faster prediction
- Automatic decision-making
- Improved customer experience

5. Big Data Technology

Modern flight prediction systems use Big Data to process large datasets collected from airline bookings, customer behavior, and travel websites.

This helps companies:

- Understand customer preferences
- Predict price trends
- Improve airline revenue management

CHAPTER 2. REVIEW OF LITERATURE

Many researchers and data analysts have worked on flight price prediction using Machine Learning techniques. The main objective of these studies is to predict airline ticket prices accurately based on different factors such as airline, source city, destination city, departure time, arrival time, duration, stops, class type, and booking date.

Previous studies show that flight ticket prices change dynamically depending on travel demand, airline services, seasonal trends, and booking timing. Researchers have applied different regression and machine learning algorithms such as Linear Regression, Decision Tree, Random Forest, Support Vector Machine, and XGBoost to improve prediction accuracy. Among these methods, ensemble models like Random Forest often produced better results because they can handle complex relationships in the dataset effectively.

Several datasets available on [Kaggle](#) have been widely used for flight fare prediction projects. These datasets contain both categorical and numerical features, making them suitable for data pre-processing, feature engineering, exploratory data analysis, and regression modelling. Researchers observed that factors such as flight duration, number of stops, class type, and days left before departure strongly influence ticket prices.

Some research papers also highlighted that early booking generally results in lower ticket prices, while last-minute bookings are usually more expensive.

In this project, the Kaggle Flight Price Prediction dataset is used to analyze flight pricing patterns and build a machine learning model for predicting ticket prices. The project includes data preprocessing, exploratory data analysis (EDA), feature encoding, regression modeling, and model evaluation. The study aims to understand how different flight-related features affect ticket pricing and how machine learning can help in making accurate predictions.

1. Introduction to Flight Price Prediction

Flight ticket prices change frequently based on several factors such as demand, airline company, season, journey date, departure time, and destination. Researchers have used data analysis and machine learning techniques to predict ticket prices accurately. Flight price prediction helps customers book tickets at lower prices and assists airlines in revenue management.

2. Previous Research on Machine Learning in Price Prediction

Many researchers have applied machine learning algorithms for predicting airline ticket prices. Algorithms such as Linear Regression, Decision Tree Regression, Random Forest, and XGBoost are commonly used for this purpose.

Studies show that machine learning models can identify patterns from historical flight data and estimate future prices with better accuracy compared to traditional statistical methods.

3. Use of Linear Regression in Prediction

Linear Regression is one of the simplest and most widely used supervised machine learning algorithms. It establishes a relationship between independent variables and the dependent variable.

In flight price prediction, variables such as:

Airline name

Source and destination

Duration

Number of stops

Departure time

Arrival time

Travel Class

are used to predict the final ticket price.

Researchers prefer Linear Regression because:

- It is simple to implement
- Easy to interpret
- Requires less computational power
- Suitable for numerical prediction problems

4. Importance of Data Preprocessing

Several studies emphasize the importance of data preprocessing before model training.

Researchers performs:

- Handling missing values
- Removing duplicates
- Feature engineering
- Encoding categorical variables
- Data normalization

These preprocessing techniques improve the performance and accuracy of prediction models.

5. Feature Engineering in Flight Datasets

Previous research indicates that feature engineering plays an important role in prediction accuracy. Researchers extracted additional features such as:

- Travel month
- Day of journey
- Peak travel season

- Weekend or weekday travel

These features help machine learning models understand pricing trends more effectively.

6. Comparison of Algorithms

Many studies compared Linear Regression with advanced machine learning algorithms. Results showed that:

Linear Regression provides good baseline accuracy

Random Forest and XGBoost often achieve higher accuracy

Complex models require more training time and computational resources

Despite lower accuracy compared to advanced models, Linear Regression remains useful because of its simplicity and interpretability.

7. Challenges Identified in Previous Studies

Researchers identified several challenges in flight price prediction:

- Dynamic changes in airline pricing
- Seasonal variations
- Sudden demand fluctuations
- Limited availability of real-time data
- Large number of influencing factors

These challenges make accurate prediction difficult and require continuous model improvement.

8. Research Gap

Most previous studies focused on achieving high prediction accuracy using complex algorithms. However, fewer studies emphasized simple and interpretable models like Linear Regression for educational and analytical purposes.

This project aims to develop a flight price prediction system using Linear Regression to understand the relationship between travel-related factors and ticket prices while maintaining simplicity and efficiency.

9. Conclusion

The literature review shows that machine learning plays an important role in airline ticket price prediction. Previous studies proved that different factors significantly influence flight prices. Linear Regression is an effective algorithm for understanding pricing relationships and building a basic prediction system. Based on earlier research, this project applies data preprocessing and regression techniques to predict flight ticket prices efficiently.

CHAPTER 3. RESEARCH OBJECTIVES AND METHODOLOGY

➤ RESEARCH OBJECTIVES

- To develop a **flight price prediction model** using **Linear Regression technique**.
- To analyze how factors such as airline, departure time, number of stops, class, duration, and days left before departure affect flight ticket prices.
- To **evaluate the performance** of the model using statistical evaluation metrics such as MAE, MSE, RMSE, and R² Score.
- To **pre-process** and transform raw flight data using techniques such as data cleaning, encoding, and feature scaling for better model performance.
- To identify the most influential factors affecting flight ticket prices through statistical analysis.
- To compare actual and predicted flight prices to measure the effectiveness of the developed model.
- To build a machine learning-based solution that can help users estimate flight prices before booking tickets.
- To improve decision-making for travellers by providing estimated ticket prices using historical flight data.
- To study how the number of days left before departure influences flight ticket prices.

➤ RESEARCH PROBLEM

Flight ticket prices change frequently due to several factors such as airline type, travel class, duration, departure time, number of stops, and remaining days before departure. Because of these frequent price variations, passengers often face difficulty in identifying the best time to book tickets at affordable prices. Therefore, there is a need to develop a predictive model that

can estimate flight prices accurately based on available flight details. This study focuses on building a machine learning-based regression model to predict flight ticket prices effectively.

➤ RESEARCH DESIGN

This study follows a quantitative and predictive research design. The research is based on historical flight data collected from an online dataset. Different pre-processing techniques such as data cleaning, feature encoding, and feature scaling were applied to prepare the data for analysis. A Multiple Linear Regression model was developed to analyse the relationship between independent variables and flight prices. The model performance was evaluated using statistical metrics.

➤ TYPE OF DATA USED

The study uses secondary data collected from the **Kaggle** related to flight ticket prices. The dataset contains both categorical and numerical variables such as airline, departure time, class, stops, duration, days left, and ticket price.

➤ DATA COLLECTION METHOD

The data used in this project was collected from the Kaggle online platform. The dataset was **downloaded in CSV format** and imported into Python for pre-processing and analysis. The collected data represents historical flight booking information.

➤ DATA COLLECTION INSTRUMENT

The primary instrument used for data collection and analysis was Python programming. Libraries such as **Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn and Statsmodels** were

used for data pre-processing, model building, and evaluation. **Google Colaboratory** was used as the development environment for implementing the project.

➤ SAMPLE SIZE

The dataset used for this study contains around **3 Lakhs flight records** collected from different airlines and routes. The final sample size consisted of all valid records after data pre-processing and cleaning.

➤ SAMPLING TECHNIQUE

The study used a convenience sampling technique because the dataset was readily available from the Kaggle platform. The available flight records were selected for analysis and model development.

➤ DATA ANALYSIS TOOL

The data analysis and model development were performed using Python programming language. The following tools and libraries were used:

- Pandas – for data manipulation and preprocessing
- NumPy – for numerical computations
- Matplotlib and Seaborn – for making different graphs
- Scikit-learn – for data splitting, feature scaling, and evaluation metrics
- Statsmodels – for building the Multiple Linear Regression model
- Google Colab – for coding and analysis environment

Chapter 4: Data Analysis and Results

Univariate Analysis of Categorical Features

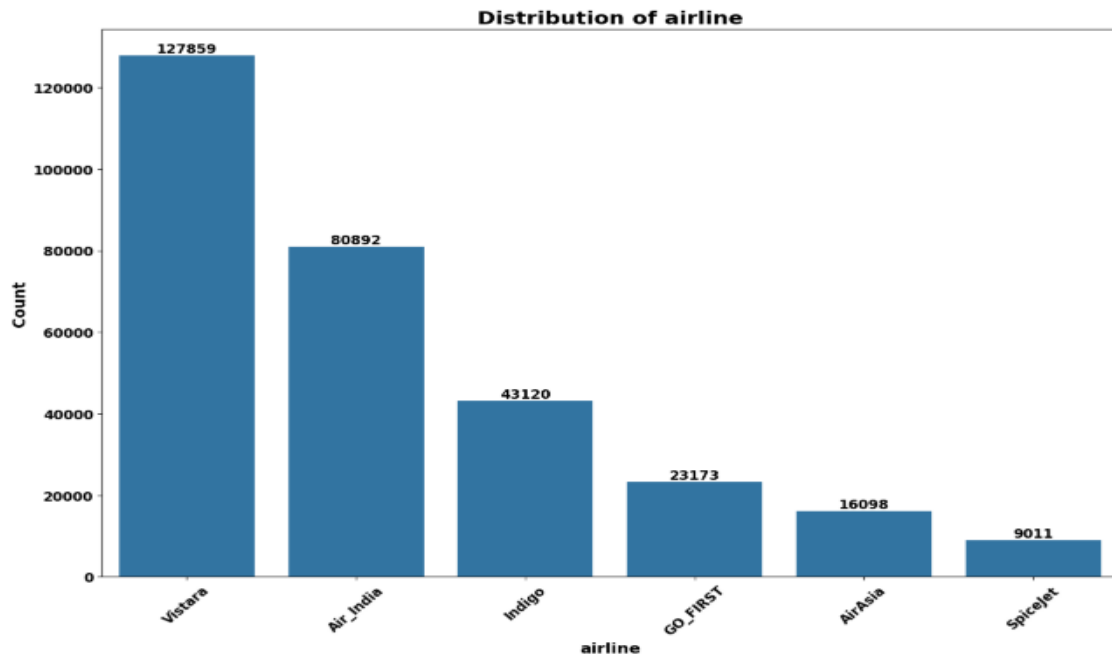


Figure 4.1: Distribution of airline

1. Distribution of Airline

The above graph “Distribution of Airline” is describing that which airline is providing more flight and which is less

Explanation:

- Some airlines have a much higher number of flights compared to others.
- According to the graph “**Vistara**” airline provides the most number of flights and “**SpiceJet**” airline provides least number of flights.
- The airline with the highest bar represents the most frequently used airline which is Vistara in the dataset.
- Airlines with smaller bars have fewer flight which is SpiceJet records.
- This indicates that airline companies do not contribute equally to the dataset.
- The airline feature may strongly affect ticket prices because different airlines provide different levels of service and pricing.

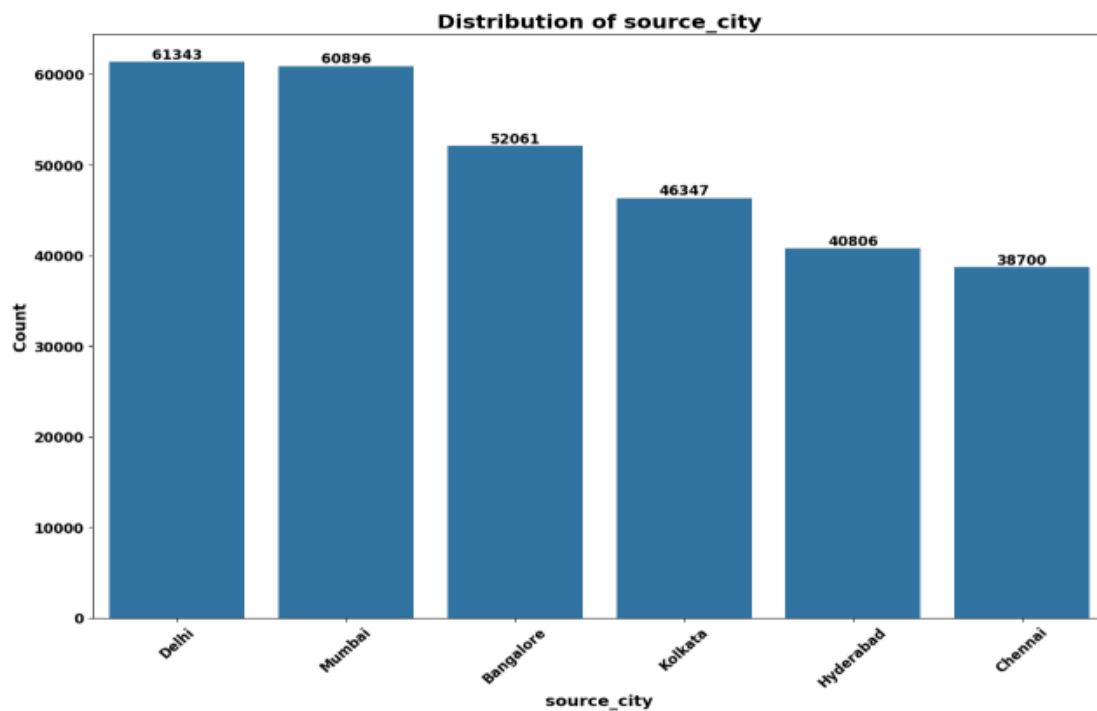


Figure 4.2: Distribution of Source City

2. Distribution of Source City

This graph represents the number of flights departing from different source cities and from city Delhi highest number of flight departing and from Chennai lowest number of flight departing

Explanation:

- Some cities have more flight departures than others.
- Cities with taller bars are major flight hubs with higher air traffic.
- According to the graph, major flight hubs with higher air traffic cities are **Delhi and Mumbai**
- The variation in source cities may influence ticket prices because travel demand differs from city to city.
- Source city is an important feature for flight price prediction.

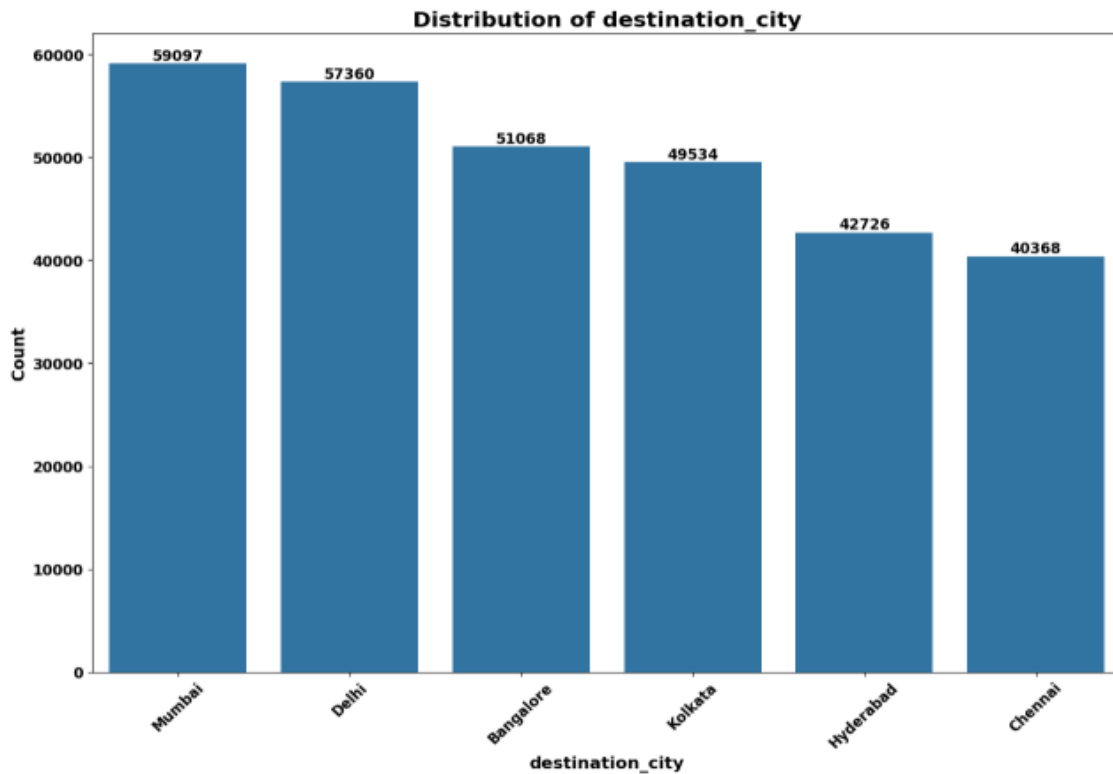


Figure 4.3: Distribution of Destination City

3. Distribution of Destination City

This graph shows the number of flights arriving at different destination cities and more flight has destination city to Mumbai and very less flight has destination city equal to Chennai.

Explanation:

- Certain destination cities have higher flight frequency. Most flights have destination cities as Mumbai and Delhi.
- Popular destination cities have more travel demand.
- Ticket prices may vary depending on destination popularity and distance.
- This feature helps the model understand travel patterns and pricing differences.

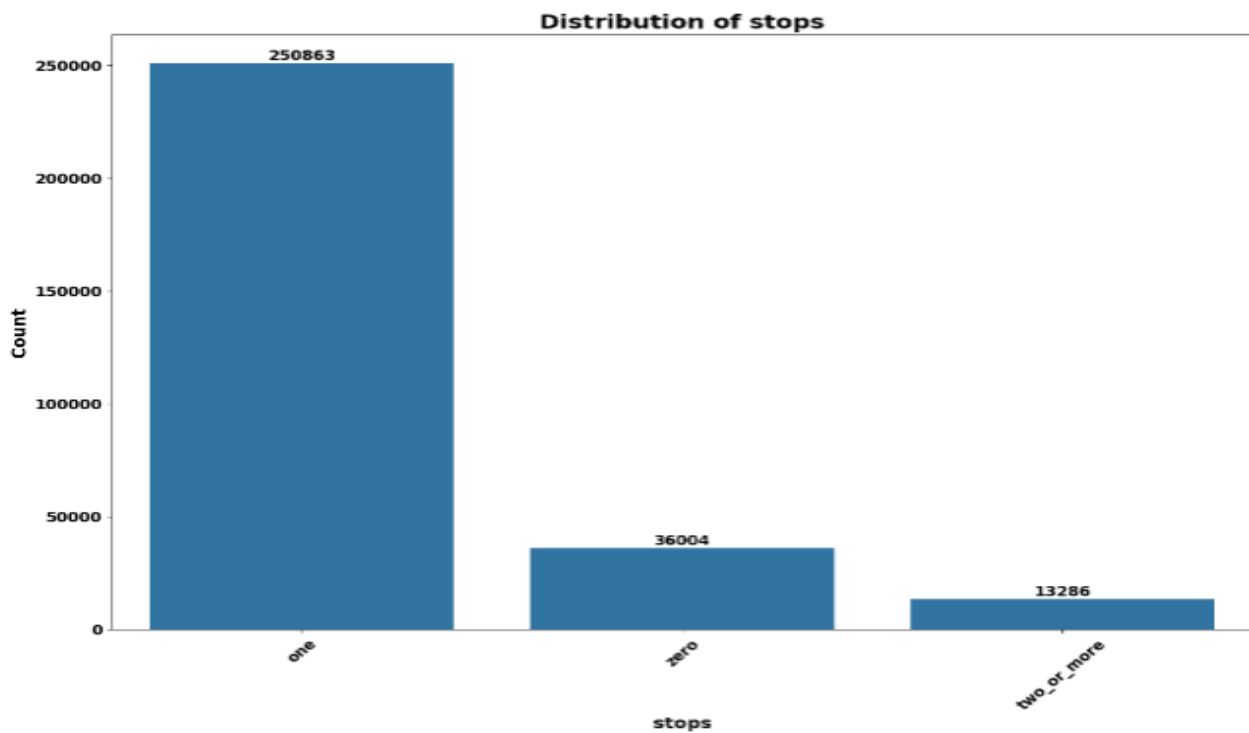


Figure 4.4: Distribution of Stops

4. Distribution of Stops

This graph shows the number of stops in flights. There are three category of stops :- one stop, zero stops, Two or more stops

Explanation:

- Most flights have one stop and then non-stop flights are there.
- According to the graph, one stop flight is more after that zero stop then two or more stops.
- Flights with two or more stops are less common because it takes more time.
- Generally, flights with more stops are expensive than direct flights.
- The number of stops significantly affects flight duration and ticket price.

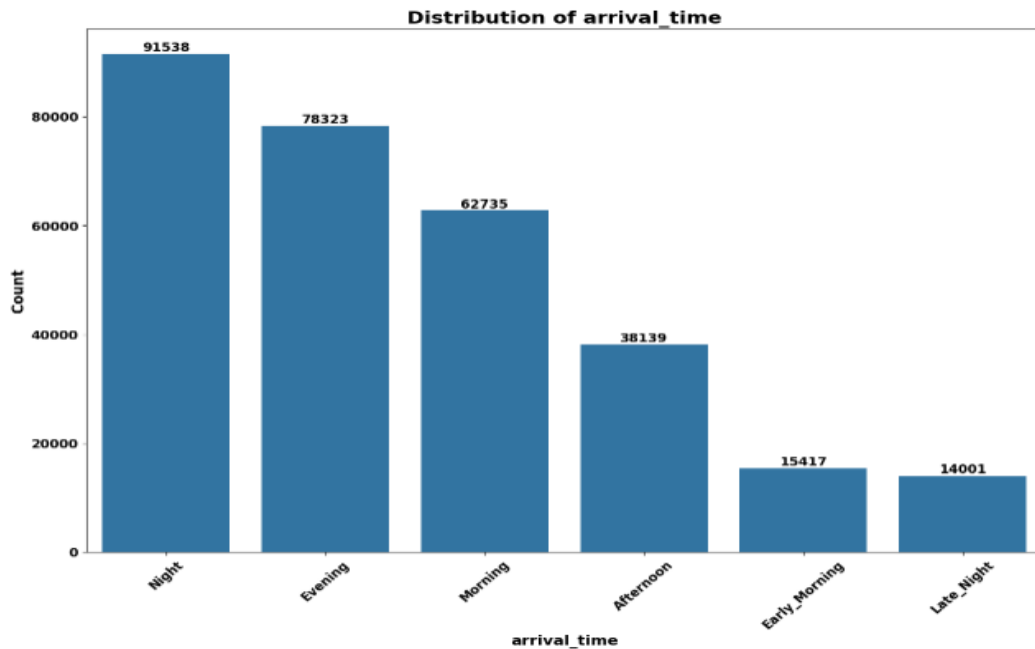


Figure 4.5: Distribution of Arrival Time

5. Distribution of Arrival Time

This graph represents the arrival time categories of flights. The arrival time is more important because passenger will prefer day time or morning time to get reached and it will easy for get the other transport also but in that graph more flight has arrival time is in night, then evening, then morning.

Explanation:

- Flights are distributed across different arrival times such as morning, afternoon, evening, night, etc.
- Night and Evening arrival flights are maximum which shows evening is the most preferred flight travel time by passengers.
- Some time periods have more flights than others.
- Arrival time may influence ticket prices because peak-hour flights are usually more expensive.
- This feature helps in understanding passenger travel preferences.

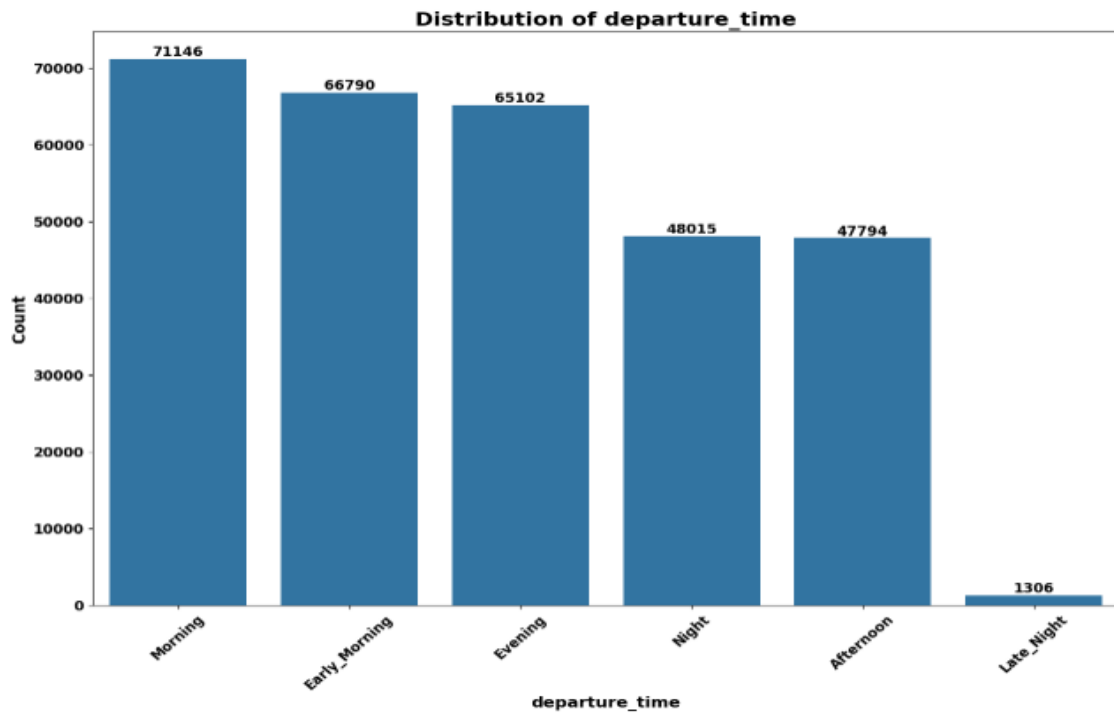


Figure 4.6: Distribution of Departure Time

6. Distribution of Departure Time

This graph shows the distribution of flights based on departure time categories such as morning, afternoon, evening, and night.

Explanation:

- Some departure time categories have more flights compared to others.
- Most passengers prefer flights during convenient time periods such as morning or evening.
- Flights departing during peak hours may have higher ticket prices because of greater demand.
- Departure time is an important feature because travel timing affects customer preference and airline pricing.
- This feature helps the model understand how departure schedules influence flight prices.

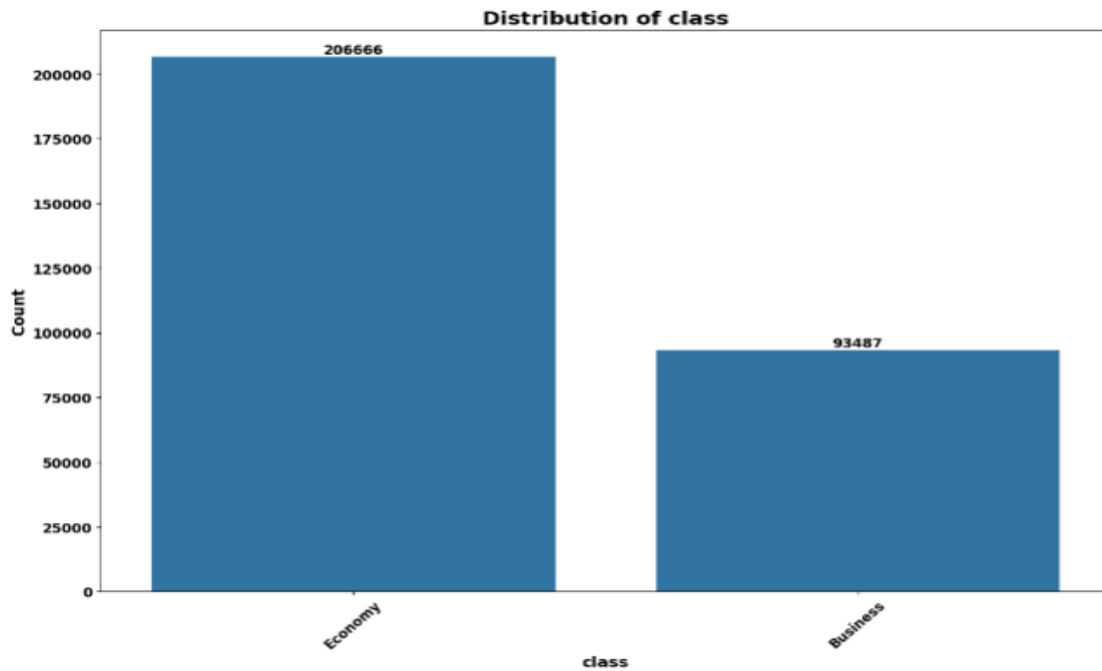


Figure 4.7: Distribution of Class

7. Distribution of Class

This graph represents the distribution of flight classes in the dataset, such as Economy and Business class.

Explanation:

- Economy class flights are usually much higher in number compared to Business class flights.
- Most passengers prefer Economy class because it is more affordable.
- Business class tickets are fewer but generally more expensive.
- Flight class has a strong impact on ticket price prediction because different classes provide different services and comfort levels.
- This feature is highly important for machine learning models while predicting ticket prices.

Conclusion

From these graphs, we can observe that:

- Flight prices depend on multiple factors such as airline, source city, destination city, stops, class, and timing.
- The dataset contains both balanced and unbalanced categories.
- Some features show strong variation, which can help machine learning models learn pricing patterns effectively.
- Exploratory Data Analysis (EDA) helps in understanding the dataset before training the prediction model.

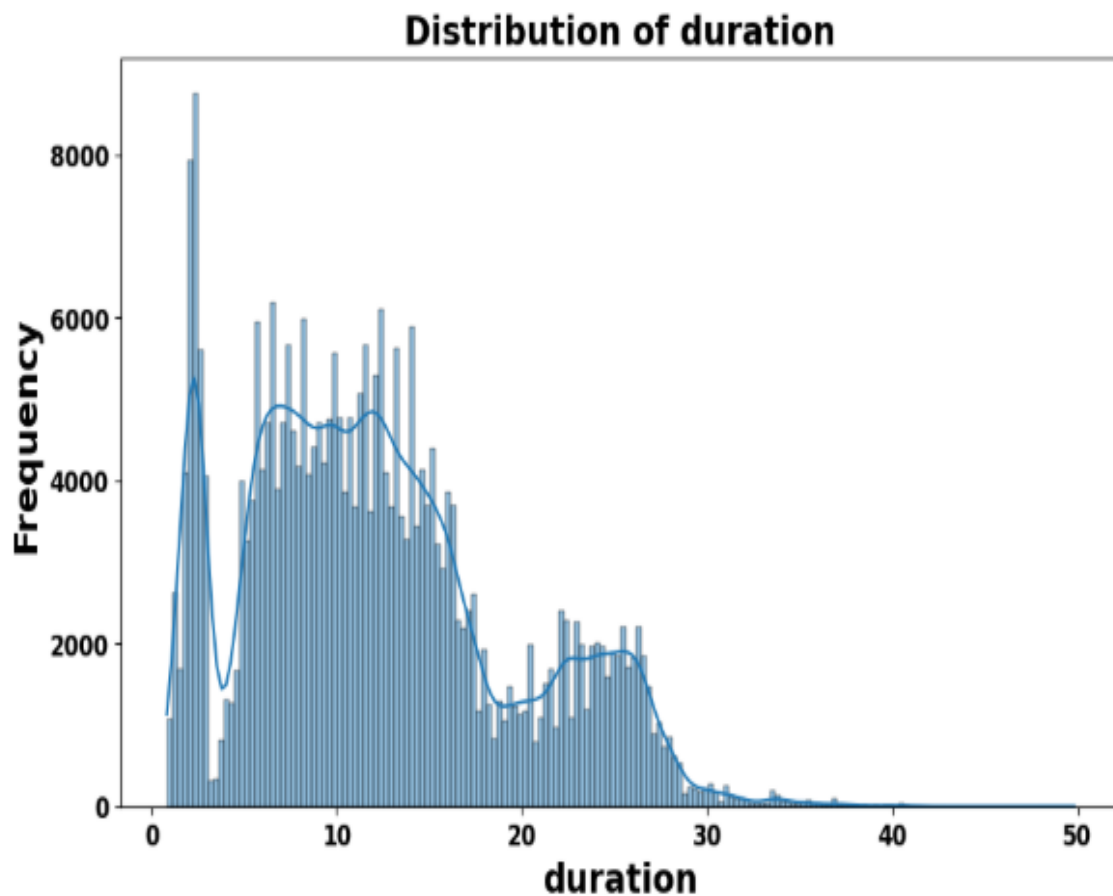


Figure 4.8: Distribution of Duration

1. Distribution of Duration

This graph shows the distribution of flight duration in the dataset.

Explanation:

- Most flights have shorter (2-3 hours with zero stops) or medium (8-10 hours with one or more stops) travel durations.
- Very long-duration flights are fewer in number.
- The graph shows that flight duration is not equally distributed.
- Flights with longer durations may have higher ticket prices because they usually cover longer distances or include multiple stops.
- Duration is an important feature for predicting flight prices.

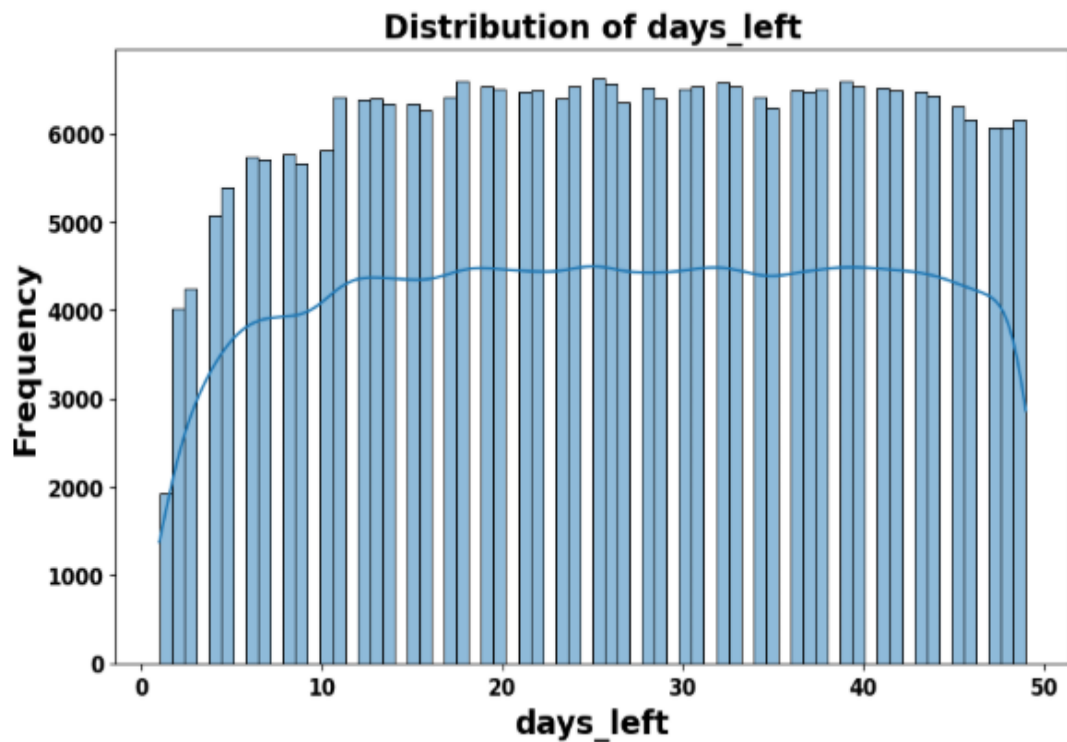


Figure 4.9: Distribution of Days Left

2. Distribution of Days Left

This graph represents the number of days left between ticket booking and journey date.

Explanation:

- The data is distributed across different booking days almost evenly.
- Ticket booking patterns are spread from early booking to last-minute booking. There are less bookings within last 10 days due to less seats availability and higher prices.
- Generally, ticket prices increase when the number of days left decreases.
- Early bookings are often cheaper, while last-minute bookings are more expensive.
- This feature plays a major role in flight price prediction.

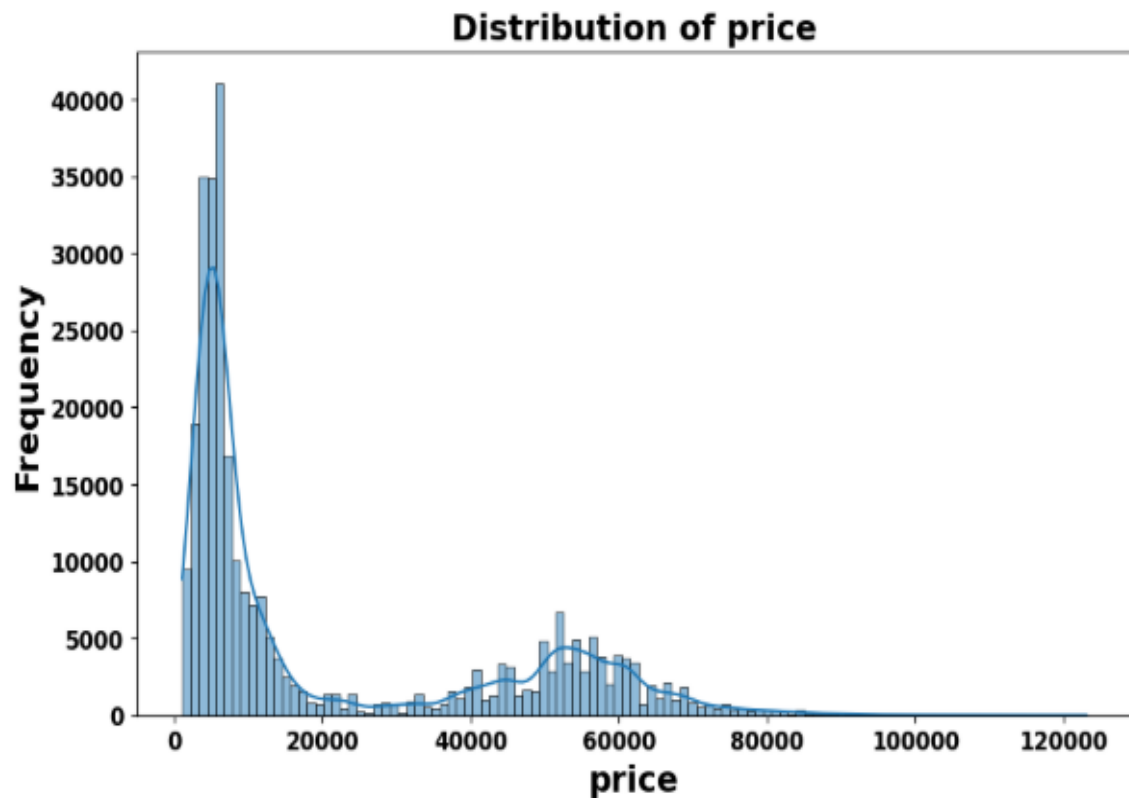


Figure 4.10: Distribution of Price

3. Distribution of Price

This graph shows the distribution of flight ticket prices.

Explanation:

- Most flight ticket prices are concentrated in the lower price range. (~10,000)
- Only a few flights have very high ticket prices.
- The graph is positively skewed (right-skewed), meaning expensive tickets are less common.
- High-price outliers may occur because of business class, premium airlines, long-distance travel, or last-minute bookings.
- Price variation indicates that multiple factors influence ticket pricing.

Overall Interpretation

From these graphs, we can conclude that:

- Flight duration, booking time, and ticket price have strong variations in the dataset.
- Ticket prices are influenced by journey duration and days left before departure.
- Most passengers book lower-priced flights, while premium-priced flights are limited.
- These insights help machine learning models understand important pricing patterns for accurate flight price prediction.

Outliers detection Using Boxplot

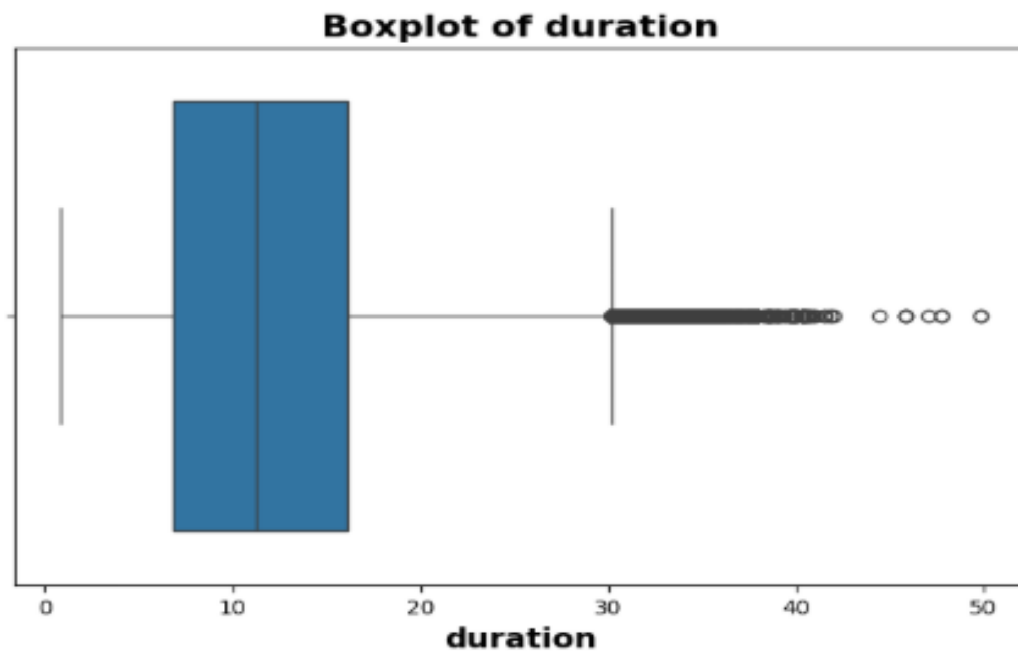


Figure 4.11: Boxplot of Duration

1. Boxplot of Duration

This boxplot represents the spread and outliers of flight duration.

Explanation:

- Most flight durations are concentrated within a certain range (inside the box).
- The middle line inside the box represents the median duration.
- There are several outliers on the higher side, indicating some flights have very long travel durations.
- These outliers may occur because of long-distance flights or multiple stops.
- The presence of outliers shows that duration varies significantly across flights.
- Duration is an important factor affecting ticket price.

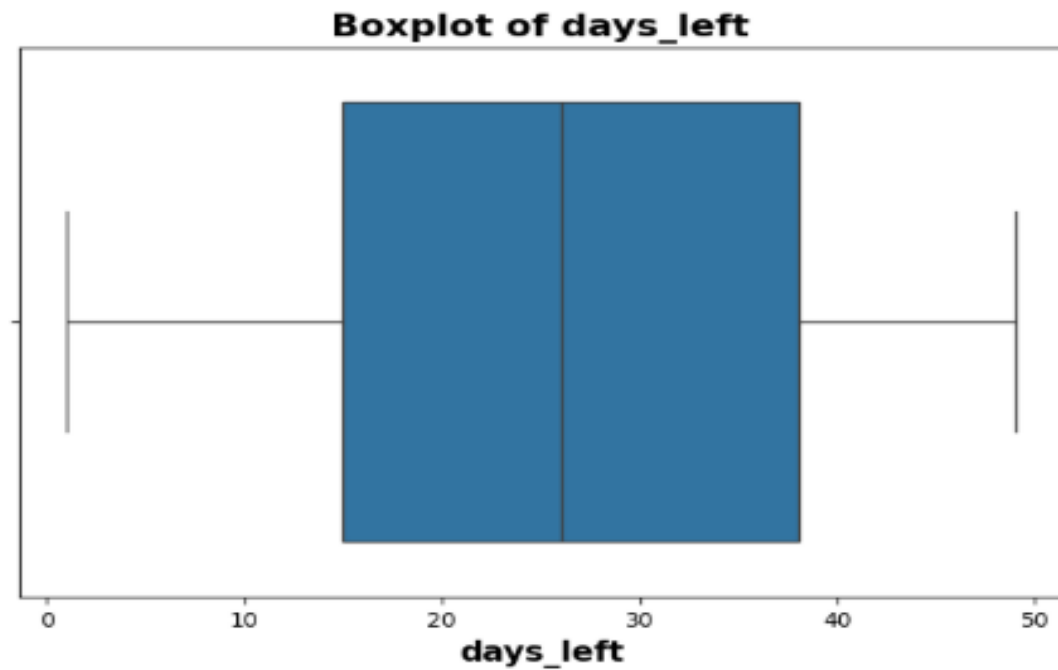


Figure 4.12: Boxplot of Days Left

2. Boxplot of Days Left

This boxplot shows the distribution of the number of days left before departure.

Explanation:

- The data is spread relatively evenly across different booking days.
- The median value is near the center, indicating balanced booking patterns.
- There are very few or no major outliers in this feature.
- This suggests that passengers book tickets at different times before travel.
- Days left is an important feature because ticket prices often increase as the departure date approaches.

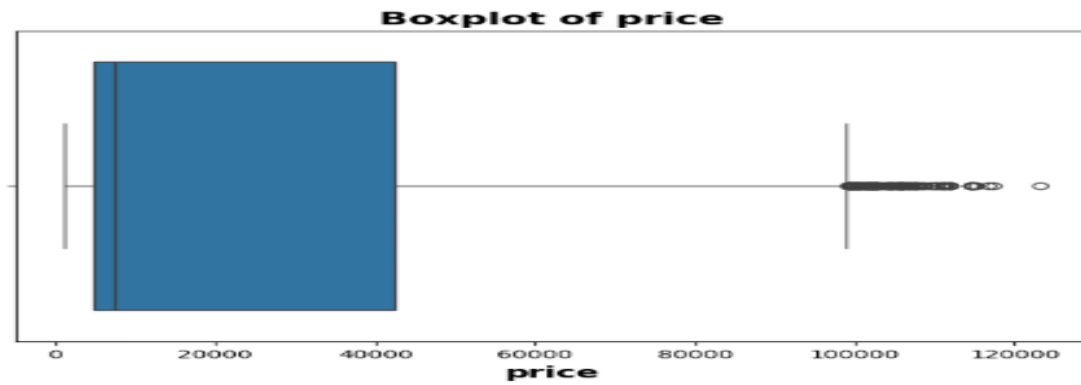


Figure 4.13: Boxplot of Price

3. Boxplot of Price

This boxplot represents the distribution and outliers of flight ticket prices.

Explanation:

- Most ticket prices are concentrated in the lower price range.
- The median price lies toward the lower side of the distribution.
- There are many outliers on the higher side, representing very expensive tickets.
- These high-price outliers may occur due to business class, premium airlines, long-distance travel, or last-minute bookings.
- The graph indicates that ticket prices are highly variable and positively skewed.
- Detecting outliers is important because they can affect machine learning model performance.

Overall Interpretation of Boxplots

From these boxplots, we can conclude that:

- Duration and price contain several outliers.
- Days_left is more uniformly distributed with fewer outliers.
- Outliers indicate extreme values that may influence model accuracy.
- Boxplots help in understanding data spread, median values, and unusual observations before model training.

Figure 4.14: Outlier Detection and Removal using IQR Method

```
# outlier detection using IQR method

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower_limit)
        (df[col] > upper_limit)

    print(f"{col}: {len(outliers)} outliers")

*** duration: 2110 outliers
    days_left: 0 outliers
    price: 123 outliers

# Removing outliers

for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower_limit) |
        (df[col] > upper_limit)
    ]

    df = df.drop(outliers.index)
```

Outlier Detection and Removal

Outliers in the numerical features (`duration`, `days_left`, and `price`) were identified and handled to ensure that extreme values do not disproportionately influence the model.

Method: Interquartile Range (IQR)

The Interquartile Range (IQR) method was employed to detect outliers. This method defines outliers as data points that fall below the first quartile (Q1) minus 1.5 times the IQR, or above the third quartile (Q3) plus 1.5 times the IQR.

- **Q1 (First Quartile):** Represents the 25th percentile of the data.
- **Q3 (Third Quartile):** Represents the 75th percentile of the data.
- **IQR (Interquartile Range):** Calculated as $Q3 - Q1$.
- **Lower Limit for Outliers:** $Q1 - (1.5 * IQR)$

- **Upper Limit for Outliers:** $Q3 + (1.5 * IQR)$

Any data point outside these calculated lower and upper limits is considered an outlier.

Outlier Identification and Impact

Initially, the following outliers were detected:

- `duration`: 2110 outliers
- `days_left`: 0 outliers
- `price`: 123 outliers

These outliers were subsequently removed from the dataset. This process led to the removal of 2232 rows, reducing the dataset from 300153 to 297921 entries. The reduction in the number of observations ensures that the statistical properties derived from the data are more robust and representative of the general trend, minimizing the skewing effect of extreme values.

After the initial removal, a re-check for outliers revealed:

- `duration`: 226 outliers (some outliers might remain due to the iterative nature of IQR outlier detection or new distribution after removal)
- `days_left`: 0 outliers
- `price`: 0 outliers

The `price` column, which is the target variable, showed successful removal of all its identified outliers, contributing to a cleaner dataset for model training.

Bivariate Analysis of Categorical Features

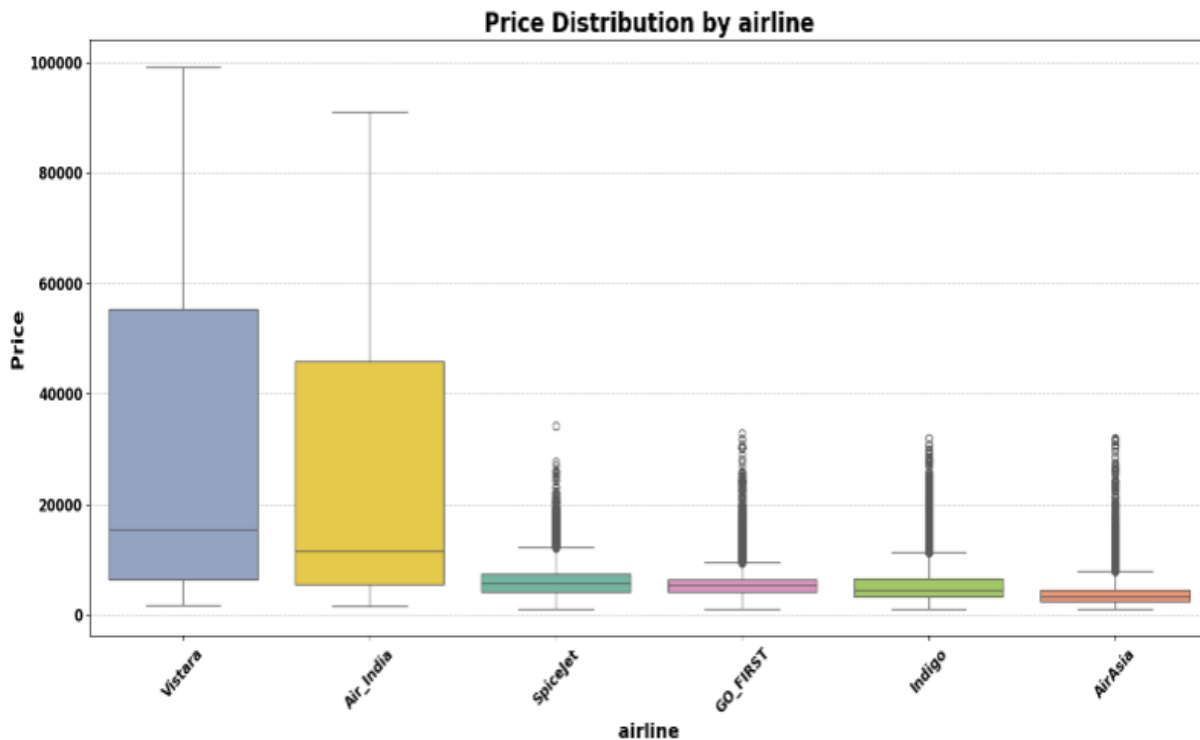


Figure 4.15: Price Distribution by Airline

1. Price Distribution by Airline

This boxplot shows how ticket prices vary across different airlines.

Explanation:

- Some airlines have higher median ticket prices compared to others.
- Airlines such as Vistara and Air India show higher price ranges.
- Airlines like Indigo and AirAsia generally have lower ticket prices.
- Several outliers are present, indicating some unusually expensive tickets.
- This shows that airline choice strongly affects flight ticket price.
- Premium airlines usually provide better services, leading to higher prices.



Figure 4.16: Price Distribution by Source City

2. Price Distribution by Source City

This graph represents how ticket prices vary based on source cities.

Explanation:

- Ticket prices differ across different departure cities.
- Some cities show a wider spread of prices, indicating high price variation.
- Certain cities have more expensive flights due to higher demand or longer travel routes.
- Outliers indicate some very costly flights from specific cities.
- Source city plays an important role in flight price prediction.

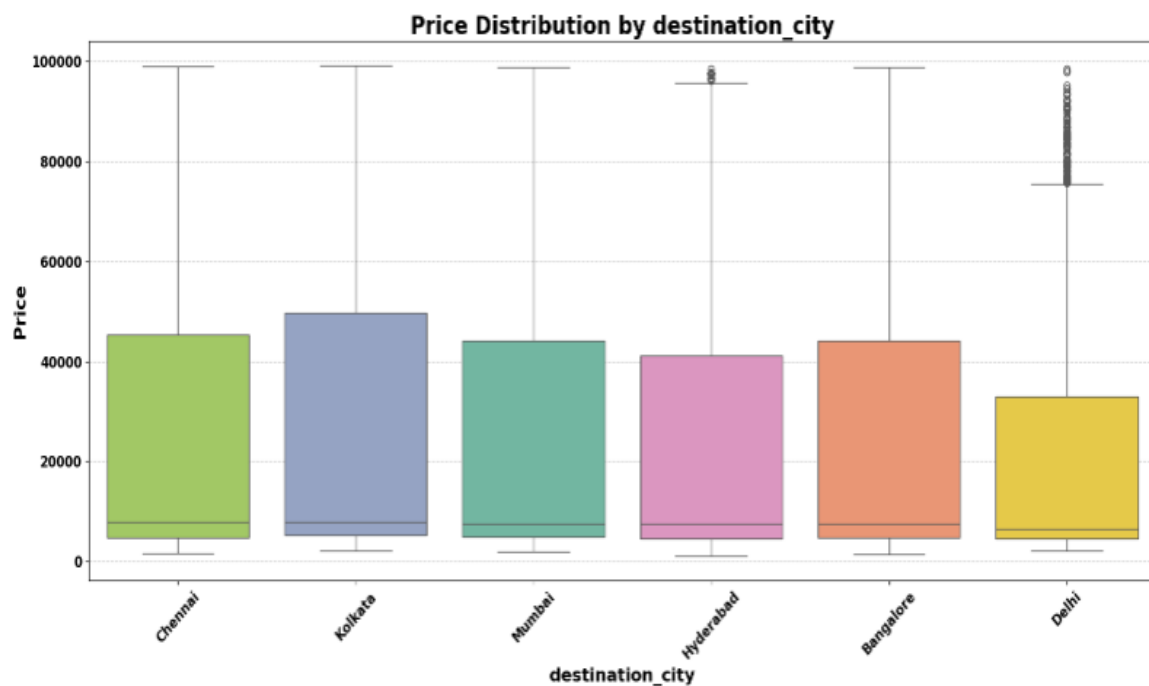


Figure 4.17: Price Distribution by Destination City

3.Price Distribution by Destination City

This graph represents the variation of ticket prices across different destination cities.

Explanation:

- Ticket prices vary depending on the destination city.
- Some cities show higher median prices due to higher travel demand or longer travel distance.
- A wider box indicates greater variation in ticket prices for that city.
- Outliers represent some unusually expensive flights to specific destinations.
- Destination city is an important factor in flight price prediction because travel routes and demand differ from city to city.

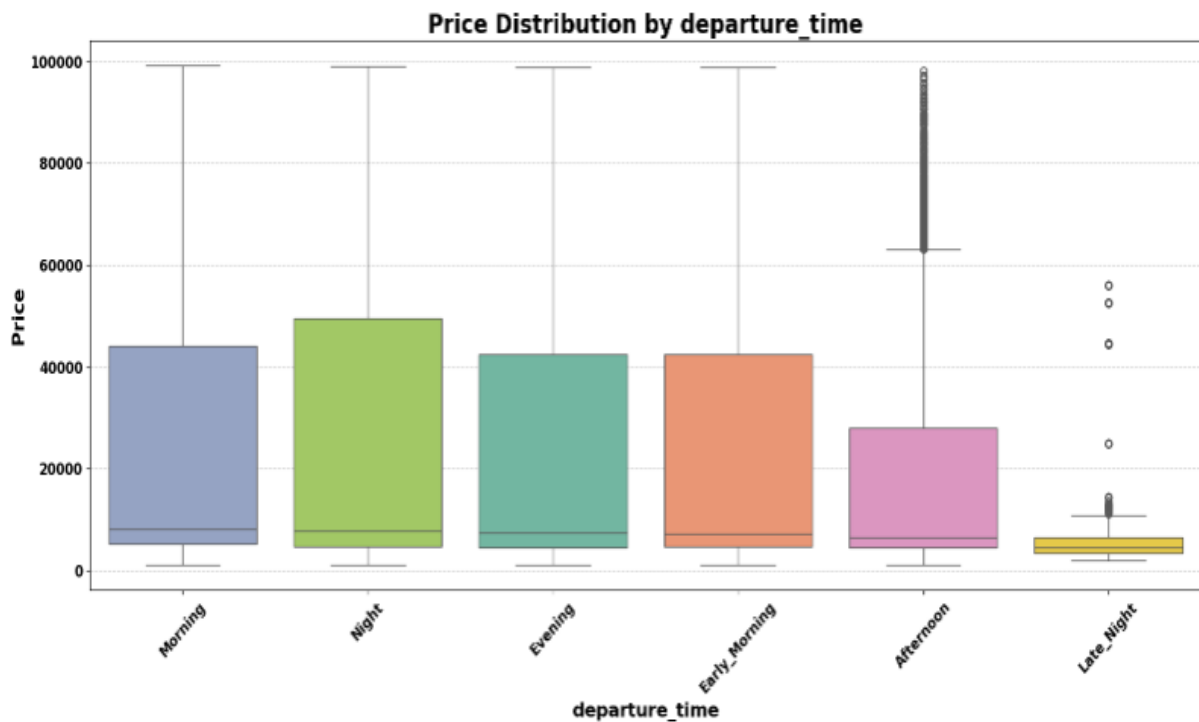


Figure 4.18: Price Distribution by Departure Time

4. Price Distribution by Departure Time

This graph shows how flight ticket prices vary according to departure time.

Explanation:

- Ticket prices vary across different departure time categories such as morning, evening, and night.
- Some departure times have higher median prices than others.
- Flights during peak travel hours are generally more expensive.
- Late-night flights appear comparatively cheaper.
- Outliers indicate some premium or last-minute flights with very high prices.
- Departure time affects passenger demand and therefore influences pricing.

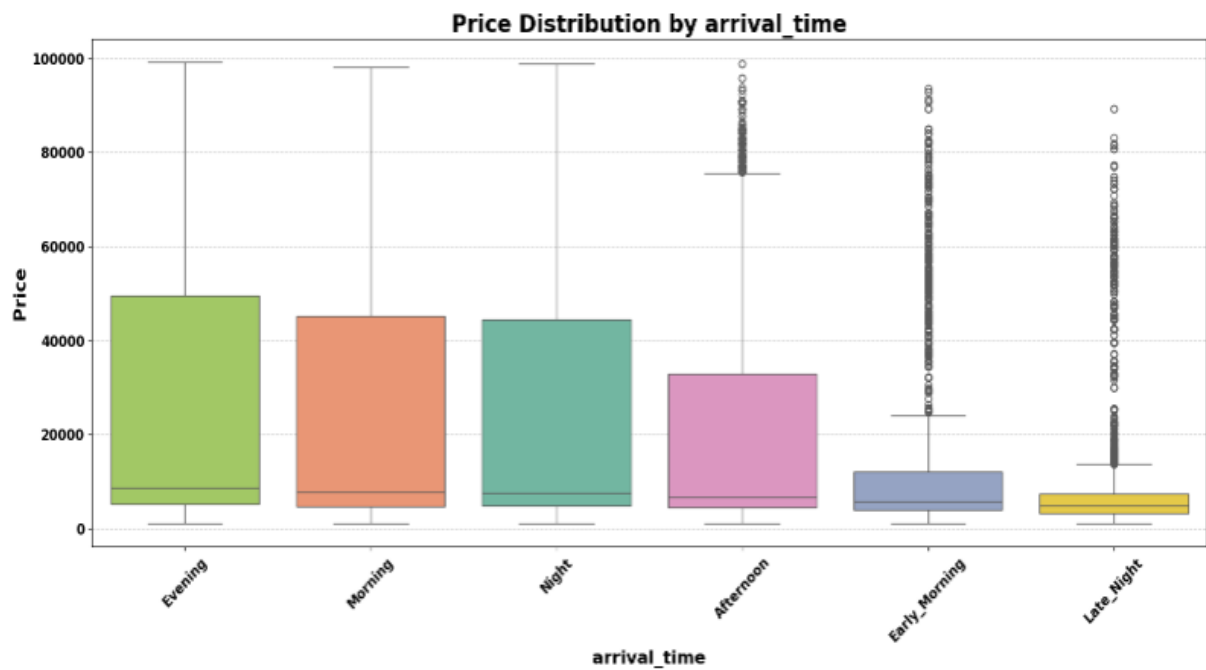


Figure 4.19: Price Distribution by Arrival Time

5. Price Distribution by Arrival Time

This boxplot shows how ticket prices vary according to different arrival times.

Explanation:

- Flight prices differ based on arrival time categories such as morning, afternoon, evening, and night.
- Some arrival times have a wider spread of prices, indicating large price variation.
- Early morning and late-night arrivals generally have lower median prices.
- Peak-time arrivals may have higher ticket prices because of greater passenger demand.
- Many outliers are visible, representing premium or expensive flights.
- Arrival time influences passenger convenience and therefore affects ticket pricing.

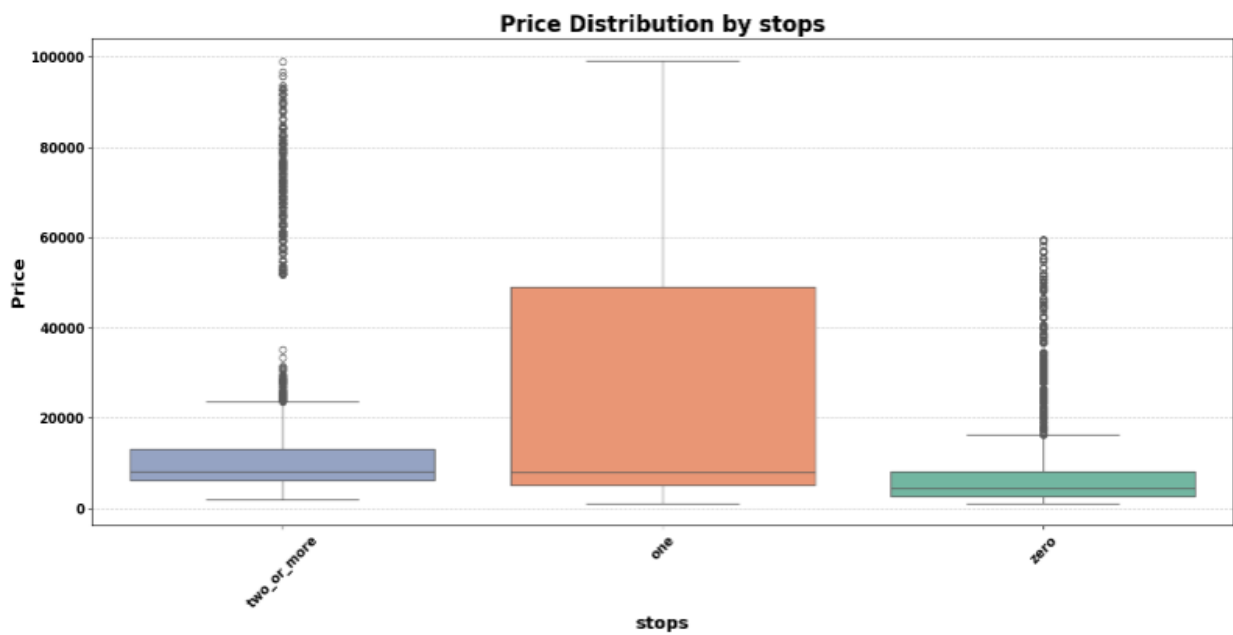


Figure 4.20: Price Distribution by Stops

6. Price Distribution by Stops

This graph represents the relationship between number of stops and ticket prices.

Explanation:

- Flights with more than one stops generally have higher ticket prices compared to one-stop and non-stop flights.
- Non-stop flights are often more convenient and also be cheaper.
- Flights with more stops are usually expensive because they take more travel time.
- Many outliers are visible, representing premium-priced tickets.
- The number of stops significantly affects both flight duration and ticket price.

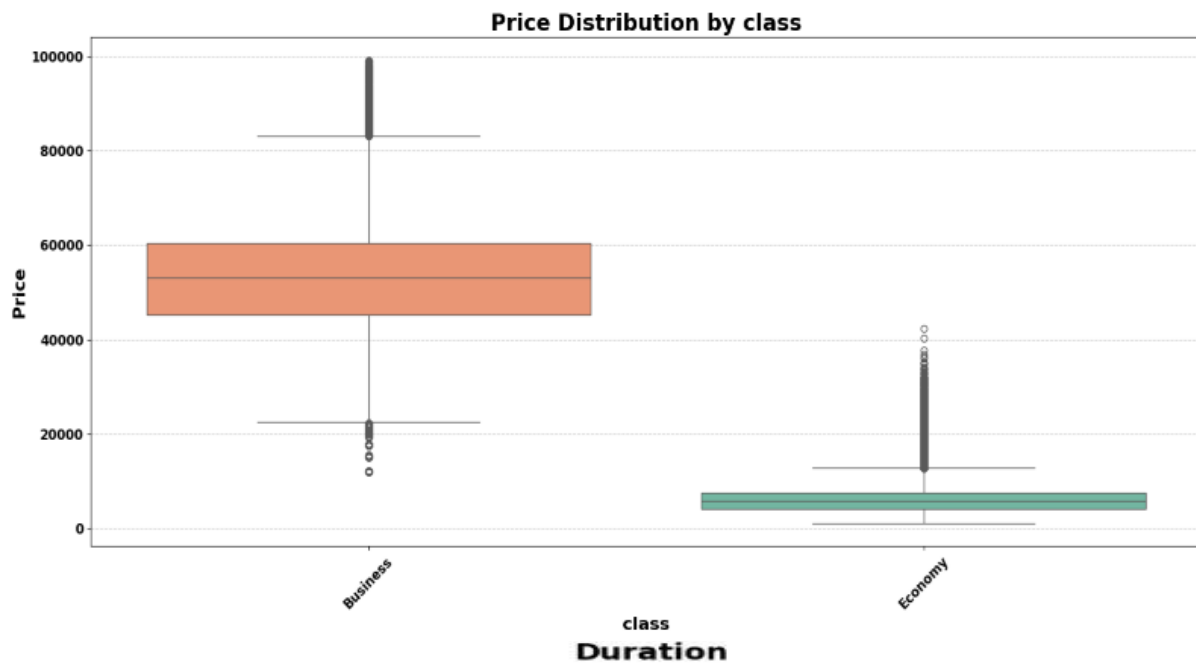


Figure 4.21: Price Distribution by Class

7. Price Distribution by Class

This graph shows the relationship between flight class and ticket prices.

Explanation:

- Business class tickets are significantly more expensive than Economy class tickets.
- The median price of Business class is much higher than Economy class.
- Economy class has lower ticket prices because it is more affordable and commonly used by passengers.
- Business class also shows a larger spread of prices, indicating greater price variation.
- Outliers are present in both classes, representing unusually high-priced tickets.
- Flight class is one of the most important features affecting ticket price prediction.

Bivariate Analysis of Numerical Features

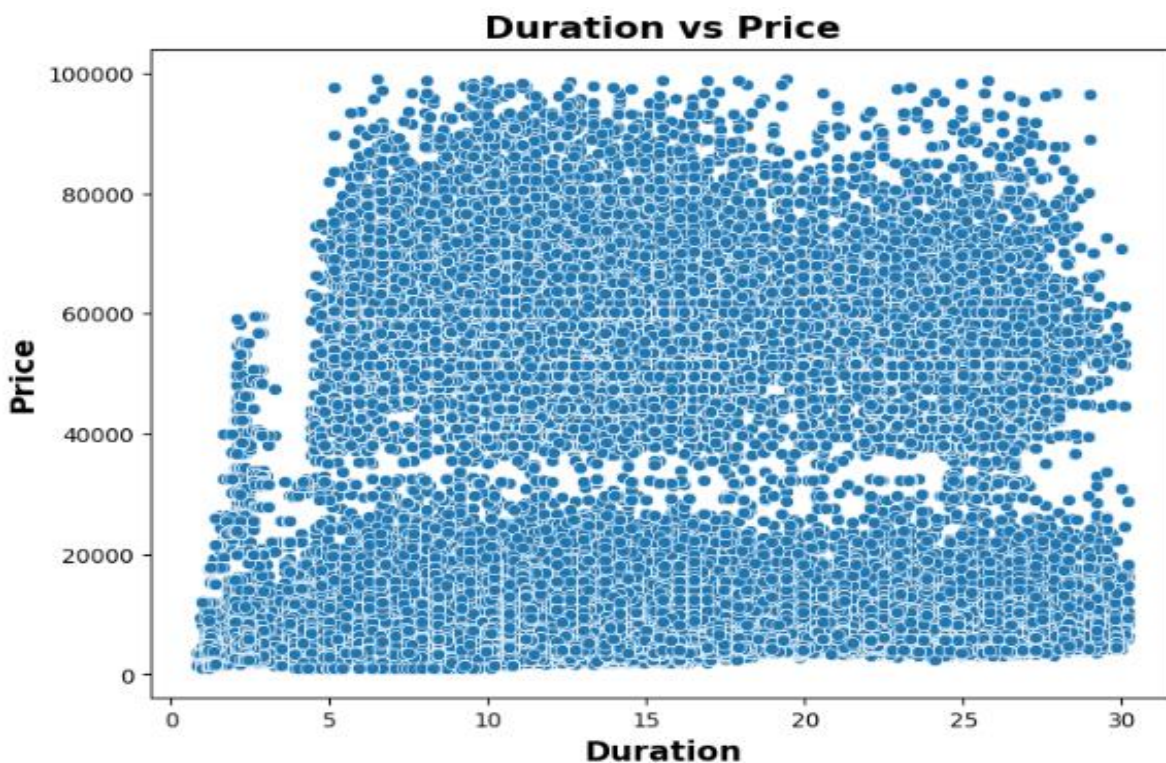


Figure 4.22: Duration vs Price

1. Duration vs Price

This scatter plot shows the relationship between flight duration and ticket price.

Explanation:

- As flight duration increases, ticket prices generally tend to increase.
- Longer flights are usually more expensive because they cover greater distances or may include premium services.
- The data points are widely scattered, which indicates that duration alone does not determine the ticket price.
- Some short-duration flights also have high prices, possibly due to premium airlines or business class tickets.
- The graph suggests a moderate relationship between duration and price.
- Duration is an important feature for flight price prediction.



Figure 4.23: Days Left vs Price

2. Days Left vs Price

This scatter plot represents the relationship between the number of days left before departure and ticket price.

Explanation:

- Ticket prices vary across all booking days.
- In general, flights booked closer to the departure date tend to have higher prices.
- Early bookings may have comparatively lower ticket prices.
- The scatter plot shows that price fluctuations occur regardless of booking day, indicating that other factors also influence pricing.
- The distribution of points suggests that days_left has an impact on ticket price prediction.
- This feature helps understand customer booking behavior.

Overall Interpretation

From these scatter plots, we can conclude that:

- Flight duration and booking timing both influence ticket prices.
- Duration shows a positive relationship with price, where longer flights often cost more.
- Ticket prices fluctuate based on days left before departure.
- Other factors such as airline, class, stops, and route also affect pricing.
- Scatter plots help identify relationships and trends between numerical variables in the dataset.

Multi-Variate Analysis

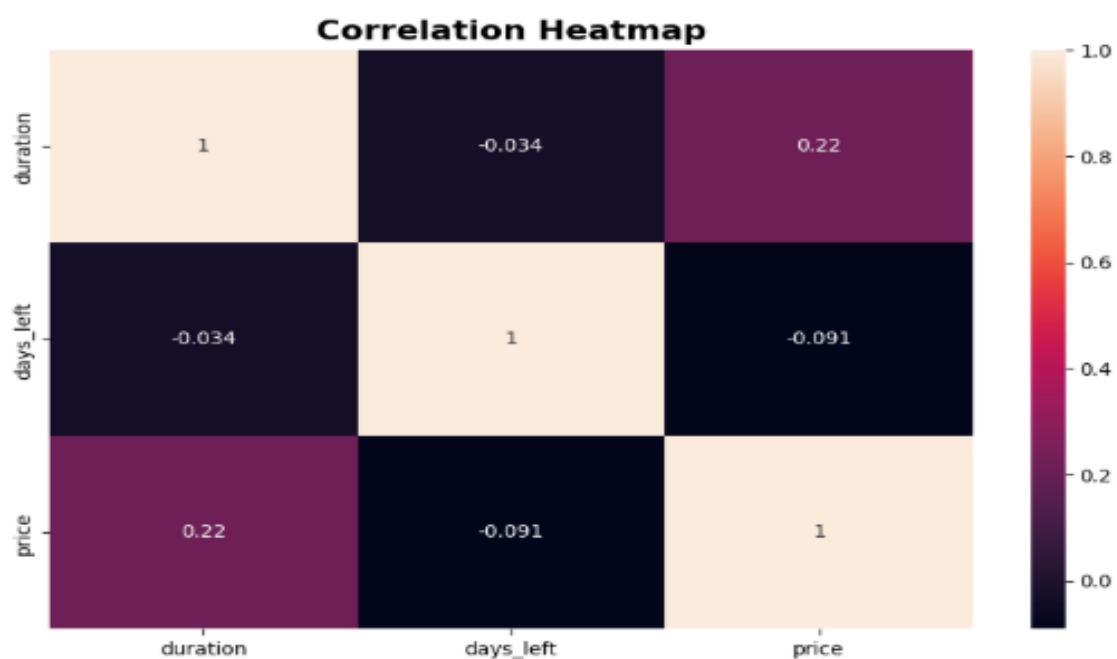


Figure 4.24: Correlation Heatmap Analysis

Correlation Heatmap Analysis

This heatmap shows the correlation between numerical features such as **duration**, **days_left**, and **price**.

Explanation:

- Correlation values range from **-1 to +1**.
 - **+1** indicates a strong positive relationship.
 - **-1** indicates a strong negative relationship.
 - **0** indicates no strong relationship.

1. Duration vs Price (0.22)

Explanation:

- The correlation value between duration and price is **0.22**.
- This indicates a **positive relationship** between flight duration and ticket price.
- As flight duration increases, ticket prices tend to increase slightly.
- Longer flights are usually more expensive because they cover longer distances or may include premium services.
- However, the correlation is not very strong, which means other factors also affect ticket price.

2. Days Left vs Price (-0.091)

Explanation:

- The correlation value between days_left and price is **-0.091**.
- This shows a **slight negative relationship**.
- As the number of days left before departure decreases, ticket prices may slightly increase.
- This means early booking can sometimes result in cheaper ticket prices.
- The correlation is weak, indicating that days_left alone does not strongly determine ticket price.

3. Duration vs Days Left (-0.034)

Explanation:

- The correlation value between duration and days_left is **-0.034**.
- This indicates almost **no relationship** between these two features.
- Flight duration does not depend on how early or late the ticket is booked.

Overall Interpretation

From the correlation heatmap, we can conclude that:

- Duration has a small positive effect on flight price.
- Days_left has a slight negative relationship with price.
- No very strong correlation exists among the numerical features.
- Flight prices are influenced by multiple factors such as airline, class, stops, source city, destination city, duration, and booking timing.

Conclusion:-

- Prices not affected much by source and destination columns. The effect of these is captured by duration. So, we can remove source and destination from the model.
- With departure and duration, we can find arrival time. So, it is redundant column. We can remove arrival time from the model.
- Also, flights we can remove. And take airlines in general.
- So final categorical columns - airline, departure_time, stops, class

Linear Regression

```
# Import libraries
import pandas as pd
import statsmodels.api as sm

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Columns
cat_cols = ['airline', 'departure_time', 'stops', 'class']
num_cols = ['duration', 'days_left']

# Features and target
X = df[cat_cols + num_cols]
y = df['price']

# One-hot encoding
X_encoded = pd.get_dummies(X, columns=cat_cols, drop_first=True)

# Convert all columns to float
X_encoded = X_encoded.astype(float)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_encoded,
    y,
    test_size=0.2,
    random_state=42
)

# Scaling
scaler = StandardScaler()

X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols] = scaler.transform(X_test[num_cols])

# Add constant to both train and test
X_train_sm = sm.add_constant(X_train)
X_test_sm = sm.add_constant(X_test)

# Build model
model = sm.OLS(y_train, X_train_sm).fit()

# Summary
print(model.summary())
```

```
import pandas as pd
import statsmodels.api as sm

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

Explanation:

- pandas is used for handling and manipulating the dataset.
- statsmodels.api is used to build the Linear Regression model and generate a detailed statistical summary.
- train_test_split helps divide the dataset into training and testing sets.
- StandardScaler is used to normalize numerical features so that all values are on a similar scale.

2. Selecting Columns

```
cat_cols = ['airline', 'departure_time', 'stops', 'class']
num_cols = ['duration', 'days_left']
```

Explanation:

The dataset contains two types of features:

Categorical Columns

The following categorical features are selected for the model:

- airline
- departure_time
- stops
- class

Numerical Columns

These following numerical features are selected for the model:

- duration
- days_left

Separating them helps in pre-processing and encoding.

3. Defining Features and Target Variable

```
X = df[cat_cols + num_cols]
y = df['price']
```

Explanation:

- X contains all input features used for prediction.
- y contains the target variable, which is the flight ticket price.

In Simple Words:

The model learns the relationship between flight details and ticket price.

4. One-Hot Encoding for categorical variables

```
X_encoded = pd.get_dummies(X, columns=cat_cols, drop_first=True)
```

Explanation:

Machine Learning models cannot directly understand text data.

Therefore, **categorical columns** are converted into numerical format using **One-Hot Encoding**.

Example:

If airline values are Air India, Indigo and Vistara. Their One-Hot Encoding would be:

Table 4.1: One-Hot Encoding of Airline Feature

Airline	Airline_Indigo	Airline_Vistara
Air India	0	0
Indigo	1	0
Vistara	0	1

drop_first=True removes one column to avoid multicollinearity (dummy variable trap).

5. Converting Data Type to Float

```
X_encoded = X_encoded.astype(float)
```

Explanation:

All feature columns are converted into float datatype because regression models require numerical input values.

6. Splitting Data into Training and Testing Sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X_encoded,
    y,
    test_size=0.2,
    random_state=42
)
```

Explanation:

The dataset is divided into:

- **80% Training Data**
- **20% Testing Data**

Purpose:

- Training data is used to train the model.
- Testing data is used to evaluate model performance.

Parameters:

- `test_size=0.2` →
20% data used for testing.
- `random_state=42` →
ensures reproducible results.

7. Standard Scaling for numerical columns

```
scaler = StandardScaler()

X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols] = scaler.transform(X_test[num_cols])
```

Explanation:

Feature scaling standardizes numerical columns so they have:

- Mean = 0
- Standard deviation = 1

Why Scaling is Important?

Numerical columns may have different ranges.

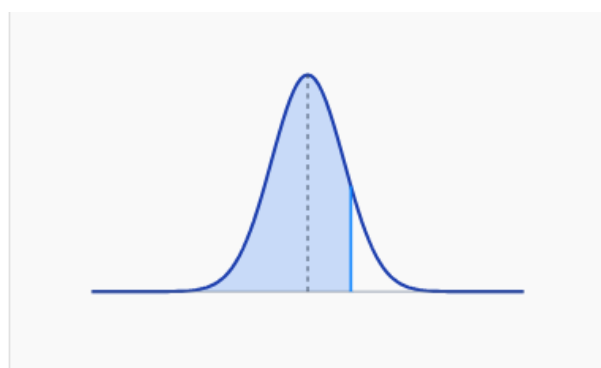
Example:

- duration → may range from 1 to 20 hours
- days_left → may range from 1 to 50 days

Scaling prevents larger values from dominating the model.

Standardization Formula

$$z = \frac{x - \mu}{\sigma}$$



Where:

- x = original value
- μ = mean
- σ = standard deviation

8. Adding Constant Value

```
X_train_sm = sm.add_constant(X_train)
X_test_sm = sm.add_constant(X_test)
```

Explanation:

A constant column is added to represent the intercept (β_0) in the regression equation.

Linear Regression Equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

Where:

- y = predicted flight price
- β_0 = constant
- $\beta_1, \beta_2 \dots$ = coefficients
- $x_1, x_2 \dots$ = feature values

9. Building the Linear Regression Model

```
model = sm.OLS(y_train, X_train_sm).fit()
```

Explanation:

- OLS stands for **Ordinary Least Squares** regression.
- The model tries to find the best-fit line that minimizes prediction error.

Purpose:

To learn the relationship between input features and flight prices.

10. Displaying Model Summary

```
print(model.summary())
```

Explanation:

This generates a detailed statistical report of the regression model.

Important Metrics in Summary:

- **R-squared** → Measures model accuracy.
- **Adjusted R-squared** → Accuracy adjusted for number of features.
- **P-value** → Checks feature significance.
- **Coefficient values** → Shows feature impact on price.

Prediction

11. Predicting Flight Prices

```
y_pred = model.predict(X_test_sm)
```

Explanation:

The trained model predicts flight prices for test data.

Output:

Predicted ticket prices based on input features.

Model Evaluation

12. Evaluating Model Performance

```
print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", mean_squared_error(y_test, y_pred) ** 0.5)
print("R2 Score:", r2_score(y_test, y_pred))
```

Explanation:

Different evaluation metrics are used to measure prediction accuracy.

MAE (Mean Absolute Error)

```
mean_absolute_error(y_test, y_pred)
```

Formula

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Meaning:

Measures the average absolute difference between actual and predicted prices.

Lower MAE = Better Model

MSE (Mean Squared Error)

```
mean_squared_error(y_test, y_pred)
```

Formula

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Meaning:

Calculates squared prediction errors.

Large errors receive higher penalties.

Lower MSE = Better Performance

RMSE (Root Mean Squared Error)

```
mean_squared_error(y_test, y_pred) ** 0.5
```

Formula

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Meaning:

Square root of MSE.

It gives error in the same unit as ticket price.

Lower RMSE = Better Predictions

R² Score (Coefficient of Determination)

```
r2_score(y_test, y_pred)
```

Formula

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Meaning:

Shows how much variation in flight prices is explained by the model.

Range:

- 1 → Perfect prediction
- 0 → Poor prediction

Higher R² Score = Better Model

Overall Workflow of the Model

1. Import libraries
2. Select categorical and numerical features
3. Define input and target variables
4. Apply One-Hot Encoding
5. Convert data into numerical format
6. Split dataset into train and test sets
7. Scale numerical features
8. Add constant term
9. Train Linear Regression model
10. Generate model summary
11. Predict flight prices
12. Evaluate model performance using MAE, MSE, RMSE, and R^2 score

Results:

```
print(model.summary())
```

The output of Linear Regression is as follows:

Simple Explanation of OLS Regression Results

Table 4.2: Statistical Summary of OLS Regression Model

OLS Regression Results						
=====						
Dep. Variable:	price	R-squared:	0.909			
Model:	OLS	Adj. R-squared:	0.909			
Method:	Least Squares	F-statistic:	1.583e+05			
Date:	Sat, 23 May 2026	Prob (F-statistic):	0.00			
Time:	13:27:46	Log-Likelihood:	-2.4430e+06			
No. Observations:	238336	AIC:	4.886e+06			
Df Residuals:	238320	BIC:	4.886e+06			
Df Model:	15					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	4.969e+04	77.055	644.804	0.000	4.95e+04	4.98e+04
duration	306.1223	18.999	16.113	0.000	268.885	343.360
days_left	-1761.7859	14.047	-125.424	0.000	-1789.317	-1734.255
airline_Air_India	109.7986	69.742	1.574	0.115	-26.895	246.492
airline_GO_FIRST	1718.5803	79.036	21.744	0.000	1563.671	1873.490
airline_Indigo	2335.4529	71.631	32.604	0.000	2195.057	2475.848
airline_SpiceJet	2688.2422	102.087	26.333	0.000	2488.154	2888.331
airline_Vistara	4107.9276	67.175	61.152	0.000	3976.266	4239.590
departure_time_Early_Morning	743.0918	46.392	16.018	0.000	652.164	834.019
departure_time_Evening	683.2259	47.277	14.452	0.000	590.564	775.888
departure_time_Late_Night	1247.7164	213.945	5.832	0.000	828.390	1667.043
departure_time_Morning	826.1725	45.777	18.048	0.000	736.451	915.895
departure_time_Night	554.1514	50.917	10.883	0.000	454.355	653.948
stops_two_or_more	2386.0442	71.697	33.279	0.000	2245.519	2526.569
stops_zero	-8003.7621	52.138	-153.510	0.000	-8105.952	-7901.572
class_Economy	-4.492e+04	34.207	-1313.057	0.000	-4.5e+04	-4.48e+04
=====						
Omnibus:	25668.984	Durbin-Watson:	2.007			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	189695.548			
Skew:	0.247	Prob(JB):	0.00			
Kurtosis:	7.343	Cond. No.	21.6			
=====						

This output shows the performance of the **Linear Regression (OLS)** model used for predicting flight ticket prices.

1. R-squared = 0.909

Explanation:

- R-squared tells how well the model explains the data.
- Value ranges from 0 to 1.

Interpretation:

- 0.909 means the model explains about **90.9%** of the variation in flight prices.
- **This is a very good result.**
- The model is performing well for prediction.

2. Adjusted R-squared = 0.909

Explanation:

- Adjusted R-squared is a more accurate version of R-squared.
- It considers the number of features used in the model.

Interpretation:

- Since it is also 0.909, it means **the selected features are useful and not adding unnecessary complexity.**

3. Prob (F-statistic) = 0.00

Explanation:

- This checks whether the overall model is statistically significant.

Interpretation:

- Since the value is $0.00 < 0.05$, **the model is statistically significant.**
- This means the input features are useful for predicting flight prices.

4. Important Features Affecting Price

The coefficients (coef) show how features affect flight prices.

Duration → Positive Effect

duration = 306.1223

Meaning:

- **If flight duration increases, ticket price also increases.**
- Longer flights usually cost more.

Days Left → Negative Effect

days_left = -1761.7859

Meaning:

- **As the number of days left increases, ticket price decreases.**
- Early booking is generally cheaper.

5. Airline Impact

Different airlines affect prices differently.

Examples:

Table 4.3: Airline-wise Effect on Flight Prices

Airline	Effect on Price
Vistara	Higher prices
SpiceJet	Moderate increase
Indigo	Moderate increase
GO_FIRST	Slight increase

Interpretation:

- Premium airlines usually have higher ticket prices.
-

6. Departure Time Impact

Examples:

Table 4.4: Impact of Departure Time on Ticket Prices

Departure Time	Effect
Late Night	Higher increase
Morning	Moderate increase
Evening	Moderate increase

Interpretation:

- Ticket prices vary depending on travel timing.
 - Certain time slots are more expensive because of demand.
-

7. Stops Impact

stops_two_or_more = Positive

2386.0442

Meaning:

- Flights with multiple stops have higher price than flights having one stop.
-

stops_zero = Negative

-8003.7621

Meaning:

- **Flights with zero stops have lower price than flights having one stop.**
-

8. Class Impact

class_Economy = -4.492e+04

Meaning:

- **Economy class tickets are much cheaper than Business class tickets.**
 - Class is one of the strongest factors affecting price.
-

9. P-value ($P > |t|$)

Explanation:

- P-value checks whether a feature is important.

Interpretation:

- Most features have:

P-values = 0.000

- **This means they are statistically significant and important for prediction.**

10. Durbin-Watson = 2.007

Explanation:

- Checks correlation in residual errors.

Interpretation:

- **A value close to 2 is good.**
- **It means there is no strong autocorrelation problem.**

Final Conclusion

The Linear Regression model performs very well with about **91% accuracy ($R^2 = 0.909$)**.

Features such as:

- duration
- days_left
- airline
- departure_time
- stops
- class

significantly affect flight ticket prices.

Among all features:

- **Flight class has the strongest impact.**
- **Longer duration increases price.**
- **Early booking reduces ticket price.**

Overall, the model is statistically strong and suitable for flight price prediction.

Test Data Results

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Predictions
y_pred = model.predict(X_test_sm)

# Evaluation
print("MAE :", mean_absolute_error(y_test, y_pred))
print("MSE :", mean_squared_error(y_test, y_pred))
print("RMSE:", mean_squared_error(y_test, y_pred) ** 0.5)
print("R2 Score:", r2_score(y_test, y_pred))
```

```
MAE : 4589.19300237184
MSE : 46785249.22025117
RMSE: 6839.974358157432
R2 Score: 0.9093030427677785
```

Explanation

These results show how well the Linear Regression model performs on the **test dataset**.

The evaluation metrics help measure the prediction accuracy of the model.

1. MAE (Mean Absolute Error)

MAE = 4589.19

Explanation:

- MAE calculates the average absolute difference between actual price and predicted price.

Meaning:

- On average, the model's prediction differs from the actual flight price by about **₹4589**.

Interpretation:

- Lower MAE is better.
- This value indicates that the model predictions are reasonably close to actual prices

2. MSE (Mean Squared Error)

$$\text{MSE} = 46785249.22$$

Explanation:

- MSE measures the average squared difference between actual and predicted values.
- Errors are squared, so larger errors get more penalty.

Meaning:

- A high MSE indicates the presence of some large prediction errors or outliers.

Interpretation:

- MSE is mainly used for mathematical evaluation of model performance.
- Lower MSE means better model accuracy.

3. RMSE (Root Mean Squared Error)

$$\text{RMSE} = 6839.97$$

Explanation:

- RMSE is the square root of MSE.
- It represents prediction error in the same unit as ticket price (rupees).

Meaning:

- The model prediction differs from the actual ticket price by around **₹6840** on average.

Interpretation:

- Lower RMSE indicates better prediction performance.
- RMSE is easier to understand because it uses the original price unit.

4. R² Score (R-squared)

R2 Score = 0.9093

Explanation:

- R² score shows how much variation in ticket prices is explained by the model.

Interpretation:

- 0.9093 means the model explains about **90.93%** of the flight price variation.
- This is a very strong result.
- The model fits the data very well.

Overall Interpretation

From the test results, we can conclude that:

- The model performs very well on unseen test data.
- The R² score of **90.93%** indicates high prediction accuracy.
- MAE and RMSE values show that prediction errors are reasonably low.
- The model can successfully predict flight ticket prices based on the selected features.

Final Conclusion

The Linear Regression model achieved good performance on the test dataset with high accuracy and acceptable error values. The model is effective for predicting flight prices and can capture the relationship between flight features and ticket prices successfully.

CHAPTER 5: FINDINGS AND CONCLUSION

Findings

The study successfully developed a Multiple Linear Regression model for predicting flight ticket prices using various flight-related factors such as airline, departure time, number of stops, class, duration, and days left before departure. Data preprocessing techniques including data cleaning, one-hot encoding, and feature scaling improved the quality of the dataset and helped in building an effective prediction model. The model was trained and tested using historical flight data collected from the Kaggle dataset.

The analysis showed that flight duration, class type, airline, and days left before departure had a significant impact on ticket prices. It was observed that ticket prices generally increased as the departure date became closer. Flights with business class tickets and fewer stops were comparatively more expensive than economy class and multi-stop flights respectively. Certain airlines also showed higher average ticket prices compared to others due to differences in service quality and market demand.

The performance evaluation metrics indicated that the regression model was able to predict flight prices with good accuracy. Metrics such as MAE, MSE, RMSE, and R^2 Score were used to evaluate the model performance. The R^2 Score demonstrated that the independent variables were able to explain a large portion of the variation in flight ticket prices. The comparison between actual and predicted prices showed that the model performed effectively in estimating ticket costs.

Highlighted Findings

- Flight duration was one of the major factors influencing ticket prices.
- Business class tickets were significantly more expensive than economy class tickets.
- Ticket prices increased when the number of days left before departure decreased.
- If flight duration increases, ticket price also increases.
- Ticket prices vary depending on travel timing. Certain time slots are more expensive because of demand.
- Flights with two or more stops generally had higher prices compared to flights with Zero and One stops.
- Airline type had a strong influence on ticket pricing patterns.
- Data preprocessing techniques improved the overall model performance and prediction accuracy.
- The Linear Regression model successfully identified relationships between flight features and ticket prices.
- The evaluation metrics showed that the developed model provided reliable flight price predictions.
- The Linear Regression model performs very well with about **91% accuracy ($R^2 = 0.909$)**.

Conclusion

The Flight Price Prediction project successfully demonstrated how machine learning techniques can be used to estimate airline ticket prices based on historical flight data. Multiple Linear Regression was used as the predictive model because it effectively analyzed the relationship between multiple independent variables and the target variable, which is flight price.

The project also highlighted the importance of data preprocessing techniques such as encoding, feature scaling, and data cleaning in improving model performance. The analysis revealed that variables such as airline, class, duration, number of stops, and days left before departure play an important role in determining ticket prices.

Overall, the developed model was able to provide accurate flight price predictions and helped in understanding the factors affecting airline pricing. This study can be useful for travelers, airline companies, and travel booking platforms for estimating ticket prices and making better booking decisions.

CHAPTER 6. RECOMMENDATIONS AND LIMITATIONS OF THE STUDY

Recommendations

- Since the model achieved a high R^2 score of about 90.9%, the selected features can be considered effective for flight price prediction.
- As ticket prices increase with flight duration, **passengers may compare shorter routes or direct flights to reduce travel costs.**
- The negative relationship between days_left and price suggests that **passengers should book tickets earlier to get lower fares.**
- The travel class analysis shows that **passengers should opt for economy class tickets for cheap prices in comparison to business class tickets.**
- Airlines and travel companies can use predictive models like this to improve pricing decisions and provide better fare estimation services to customers.

Limitations of the Study

- The model is trained only on the available Kaggle dataset, so prediction accuracy depends on the quality and size of the dataset.
- The Linear Regression model may not fully capture complex non-linear relationships between flight features and ticket prices. So, advanced regression models like neural networks, random forests etc. can be used to capture the more complex relationships.
- Real-time factors such as weather conditions, fuel prices, holidays, and airline policies are not included in the study.
- The dataset may contain outliers and missing real-world travel scenarios, which can affect model performance.
- The project mainly focuses on historical data, so sudden market changes or unexpected events cannot be predicted accurately.

BIBLIOGRAPHY

Research Papers:

1. Grover, P., Kar, A. K., & Dwivedi, Y. K. (2022). Understanding artificial intelligence adoption in operations management: Insights from machine learning applications. *International Journal of Information Management Data Insights*, 2(2), 100083.
2. Wang, X., & Yeung, D. Y. (2020). Flight ticket price prediction using machine learning techniques. *Journal of Air Transport Management*, 87, 101859.
3. Singh, A., & Sharma, S. (2021). Comparative analysis of regression algorithms for flight fare prediction. *International Journal of Computer Applications*, 174(15), 20–26.
4. Khandelwal, V., & Gupta, P. (2020). Predictive analytics for airline ticket pricing using machine learning. *International Journal of Advanced Research in Computer Science*, 11(3), 45–50.

Websites:

1. <https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction/data>

Books:

1. Gareth James, Daniela Witten, Trevor Hastie, & Robert Tibshirani, *An Introduction to Statistical Learning*, Springer Publication, 2nd Edition, pp. 101–150.
2. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 3rd Edition, pp. 45–120.
3. Trevor Hastie, Robert Tibshirani, & Jerome Friedman, *The Elements of Statistical Learning*, Springer Publication, 2nd Edition, pp. 67–140.
4. Wes McKinney, *Python for Data Analysis*, O'Reilly Media, 3rd Edition, pp. 55–180.